Experiment No 9

Studying Metasploit Framework and Msfvenom

Aim

To study the Metasploit Framework utilities in Kali Linux, inspect Msfvenom help and payload listings, and observe the output messages displayed during a controlled lab exercise.

Algorithm

1. Start Kali Linux and open the Terminal.

2. Navigate to the Metasploit Framework directory and observe the dependency installation output shown in the screenshots.

3. Open Msfvenom help to review its available options and supported output formats.

4. Inspect the listed payload categories and the selected payload's option information.

5. Review the final terminal output and note any error messages, including invalid format or option errors.

6. Perform security testing only in an isolated lab and only on systems for which explicit authorization has been granted. Do not run generated files on real systems.

Output

The following screenshots are reproduced from the supplied terminal session.

Output Screenshot 1



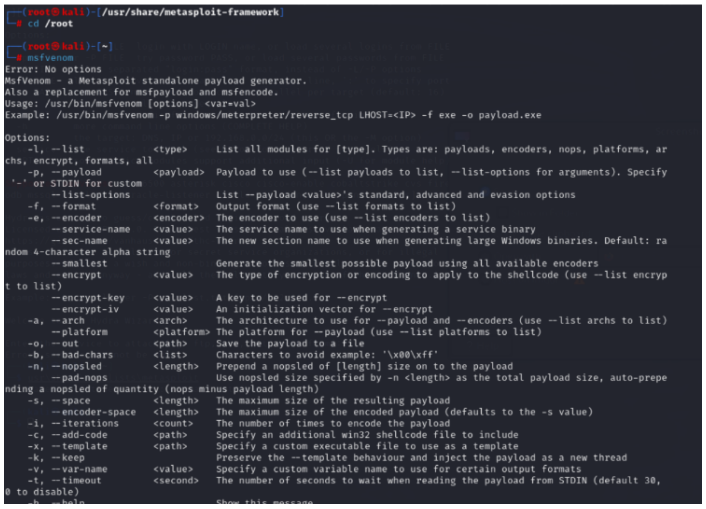
Metasploit Framework directory and Ruby dependency installation output.

Output Screenshot 2



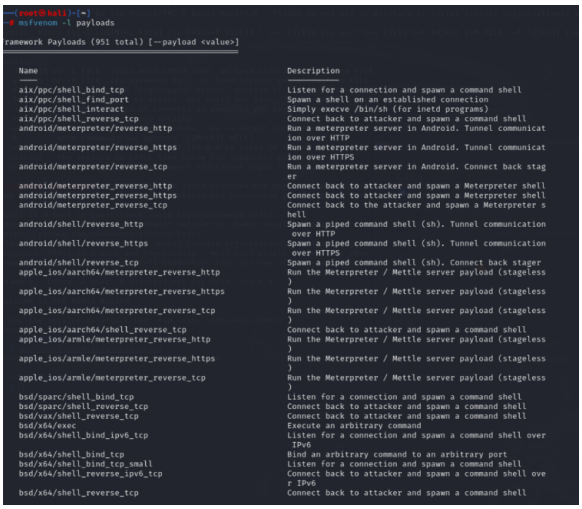
Additional dependency installation output displayed in the terminal.

Output Screenshot 3



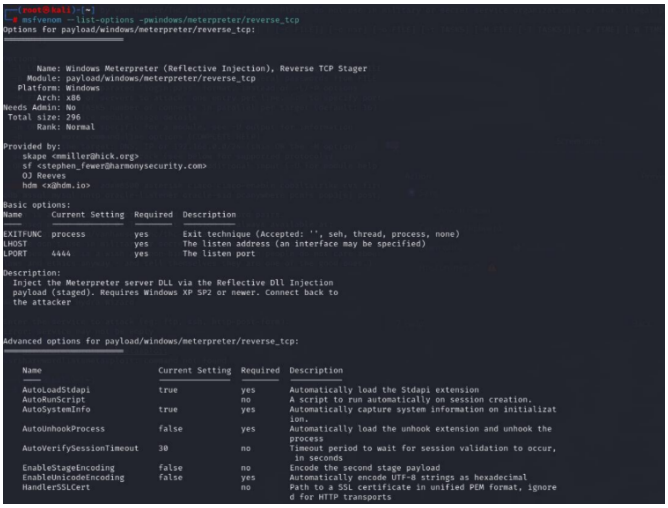
Msfvenom help screen showing usage information and supported options.

Output Screenshot 4



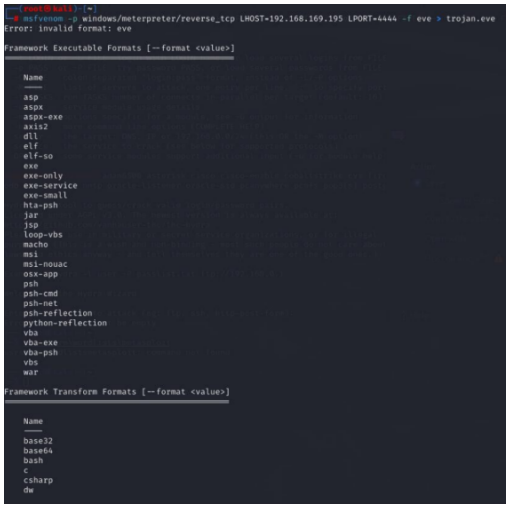
Msfvenom payload listing with example payload categories and descriptions.

Output Screenshot 5



Option details for the selected Windows Meterpreter reverse TCP payload.

Output Screenshot 6



Terminal output showing an invalid-format error and available executable output formats.

Result

Thus, the Metasploit Framework and Msfvenom command-line help were examined in Kali Linux. The screenshots show dependency installation, available options and payload information, as well as a format-related error. This experiment should be conducted only in an authorized, isolated lab environment.